

Negative-Frame Prevalence, Not Architecture, Governs False-Alarm Suppression in Edge AIoT Object Detection

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Abstract—Deep learning detectors deployed in continuous edge Artificial Intelligence of Things (AIoT) sensing face extreme temporal class imbalance: over 80% of operational frames contain no targets. Training with background-only (negative) frames suppresses false alarms, but prior work tunes negative-frame ratios for one detector architecture at a time, leaving open whether these settings transfer across designs. We present a cross-architecture study of four matched-capacity (2.3–2.6 M parameter) nano-scale detectors from the You Only Look Once (YOLO) family: YOLO11n (hybrid convolution-attention), YOLO26n (structural reparameterization), YOLO12n (area attention), and YOLOv10n (dual-label-assignment, non-maximum suppression (NMS)-free), evaluated on a 15,723-frame driver-monitoring corpus with 80.9% natural negative prevalence. Sweeping five negative-frame ratios (0%–80%), we find that false-alarm suppression trajectories diverge by up to $31.5\times$ across architectures at intermediate ratios, yet all four converge to a narrow operational band (3.14–7.08 FP/1k, false positives per 1,000 frames) at 80% negatives, with no corresponding drop in foreground recall (Spearman $\rho = +0.012$, not significant). At a fixed 40% ratio, hard-mined negatives reduce false alarms by 52.6%–67.8% for attention-based and dual-assignment models, and by 33.3% for the pure-convolutional model. These results indicate that negative-frame prevalence, not architecture or inference latency, is the primary lever for operational reliability in edge AIoT sensing.

Index Terms—edge AIoT, object detection, false-alarm suppression, negative-frame training, YOLO, hard-negative mining, driver monitoring

I. INTRODUCTION

Vision systems in continuous edge AIoT sensing must process video for hours at a time under severe foreground-background class imbalance [1], yet the safety-critical events they are built to detect are rare. In vehicular driver monitoring systems (DMS), for example, cues such as phone use, drinking, or yawning occupy only a small share of the video stream [2], while more than 80% of frames show nothing of interest.

This imbalance changes what counts as good performance for an edge detector. Mean average precision (mAP) on a curated, target-dense benchmark says little about how the same detector behaves across hours of routine footage; what matters operationally is whether the detector stays silent when there is nothing to detect. A high false-alarm rate drains battery and thermal budgets, wastes cellular bandwidth, and causes driver alert fatigue [2], [3].

Training with background-only (negative) frames is a well-established remedy for false alarms. Practitioner guidance

suggests keeping negative frames at around 0–10% of the training set [4], while studies in camera-trap ecology and open-world object detection report that omitting negative frames severely degrades false-alarm resilience [5], [6]. This guidance, however, comes from single-architecture studies: whether a negative-frame ratio or a curation strategy that works well for one detector transfers to another remains an open question. If negative-frame sensitivity depends on architecture, then detector choice and dataset curation cannot be treated as independent design decisions.

We address this gap with a focused, cross-architecture study. We fine-tune four matched-capacity (2.3–2.6 M parameter) nano-scale detectors from the YOLO family: YOLO11n [7], YOLO26n [8], YOLO12n [9], and YOLOv10n [10], spanning hybrid convolution-attention, structural reparameterization, area-attention, and dual-label-assignment (NMS-free) designs. On a 15,723-frame in-cabin DMS corpus with 80.9% natural negative prevalence, we ask:

- **RQ1 (Ratio Sensitivity):** How does the negative-frame ratio affect false-alarm suppression across architectures, and does the best-performing ratio differ by paradigm?
- **RQ2 (Curation Quality):** At a fixed reference ratio ($r_{\text{ref}} = 40\%$), does replacing random negatives with hard-mined negatives of equal count suppress false alarms further?

Our contributions are threefold. First, to our knowledge, this is among the first cross-architecture negative-frame ratio sweeps on a safety-critical AIoT dataset, spanning four detector paradigms. Second, we run a controlled hard-negative mining benchmark at matched cardinality ($r_{\text{ref}} = 40\%$), isolating sample difficulty from dataset size. Third, we pair an empirical latency benchmark with an analytical translation from FP/1k to projected hourly nuisance alerts ($\mathcal{A}_h$), showing that data curation—not inference speed—determines whether a detector is usable in continuous deployment.

II. RELATED WORK

Foreground-Background Imbalance. Class imbalance in object detection is typically addressed by loss reweighting [1], online hard-region mining [11], [12], dynamic sample assignment [13], and representation-drift tracking [14]. These techniques target candidate regions *within* positive, object-containing images. In continuous AIoT video streams, the dominant failure mode is different: entire frames contain

zero targets yet provoke spurious detections that waste edge compute, trigger unnecessary uplink transmissions, and fatigue operators [2], [3]. A unified treatment of this image-level imbalance problem, including its effect on open-world false-alarm rates, is provided by Oksuz et al. [6].

Lightweight Edge Detectors. Edge AIoT deployments require compact architectures capable of real-time inference under strict compute and thermal budgets [15]. Modern nano-scale detectors differ fundamentally in how they aggregate spatial features: hybrid convolutional-attention models (YOLO11 [7]) combine C3k2 blocks with neck-level partial self-attention (C2PSA); structural reparameterization models (YOLO26 [8]) use multi-branch RepConv during training and fuse to a single efficient kernel at inference; area-attention models (YOLO12 [9]) apply partitioned horizontal and vertical attention (A2C2f) for near-linear complexity global context; and dual-assignment models (YOLOv10 [10]) eliminate NMS via consistent one-to-one prediction heads. Evaluating these four paradigms at matched parameter count (2.3–2.6 M) isolates the effect of feature-aggregation design on background suppression.

Negative-Frame Training in AIoT. Practitioner guidelines advise including 0–10% background images [4]. Camera-trap studies show that omitting background images causes open-world failure modes [5]. In human-object interaction, negative pair scaling to 1000:1 is essential to prevent spatial spuriousness [16]. In vehicular DMS [2], [17], standard datasets [18], [19] emphasize positive-cue diversity, while deployed systems encounter vast stretches of unannotated routine driving. Critically, no prior work examines whether an optimal negative-frame ratio or hard-negative curation strategy transfers across distinct detector architectures. This paper fills that gap.

III. METHODOLOGY

A. Problem Formulation

In a continuous edge AIoT video stream $\mathcal{S} = \{X_t\}_{t=1}^T$, frames are partitioned into positive frames ($\mathcal{D}_{\text{pos}}$, containing annotated driver-safety cues) and negative frames ($\mathcal{D}_{\text{neg}}$, containing no targets). For a detector f_θ , bounding boxes predicted on frame X and filtered at operational confidence threshold $\tau \in (0, 1]$ yield the detection set $\hat{\mathcal{Y}}_\tau(X)$. Because negative frames carry no ground truth ($Y = \emptyset$), any element of $\hat{\mathcal{Y}}_\tau(X)$ is an operational false alarm.

We quantify deployment reliability via two metrics evaluated on the held-out negative test set $\mathcal{D}_{\text{test}}^{\text{neg}}$ (1,272 frames):

$$\text{FP/1k} = \frac{1000}{|\mathcal{D}_{\text{test}}^{\text{neg}}|} \sum_{X \in \mathcal{D}_{\text{test}}^{\text{neg}}} |\hat{\mathcal{Y}}_\tau(X)|, \quad (1)$$

and the projected hourly nuisance alert rate $\mathcal{A}_h$ at camera frame rate f_{FPS} (frames per second):

$$\mathcal{A}_h = 3.6 f_{\text{FPS}} p_{\text{neg}} \text{FP/1k}, \quad (2)$$

where $p_{\text{neg}} \approx 0.809$ is the operational negative prevalence of our vehicular corpus. All false-alarm metrics are evaluated at $\tau = 0.25$ and NMS intersection-over-union (IoU) threshold of

TABLE I
EVALUATED DETECTOR ARCHITECTURES (640 × 640 INPUT)

Model	Params	FLOPs	Lat.*	FPS*	Paradigm	Mechanism
YOLO11n	2.6 M	6.5 G	16.8 ms	59.6	Hybrid	C3k2 + C2PSA
YOLO26n	2.6 M	6.2 G	21.7 ms	46.1	Rep.Conv	RepConv
YOLO12n	2.6 M	7.6 G	21.3 ms	47.0	Attn	A2C2f (Area)
YOLOv10n	2.3 M	6.6 G	7.7 ms	129.2	NMS-Free	Dual-Label Head

*Measured on NVIDIA RTX 4060 Laptop GPU (PyTorch, batch=1, 640 × 640, averaged over 200 iterations after 50 warm-up iterations).

0.70. Standard accuracy metrics (mAP₅₀, mAP_{50:95}, precision P, recall R) are reported at the maximum- F_1 operating point.

B. Dataset and Partitioning

Experiments use a 15,723-frame in-cabin DMS corpus captured under diverse lighting conditions, vehicle interiors, driver subjects, and camera viewpoints, with institutional consent and de-identification. The dataset contains 3,001 positive frames annotated with verified 2D bounding boxes across four driver-cue classes: phone_use (2,437 frames, 81.2%), drinking (264 frames, 8.8%), yawning (159 frames, 5.3%), and hand_over_mouth (141 frames, 4.7%). The remaining 12,722 frames are background-only, yielding an 80.9% natural negative prevalence.

A stratified 80/10/10 split (seed=42) partitions the corpus into: (1) a held-out test set of 1,572 frames (300 positive, 1,272 negative); (2) a held-out validation set of 1,572 frames (300 positive, 1,272 negative); and (3) a training universe of 2,401 positive and 10,178 candidate negative frames. Both evaluation sets preserve the natural 80.9% negative dominance and are fixed across all experiments.

C. Detector Architectures

Table I summarizes the four nano-scale detectors benchmarked in this study. All models operate within 2.3–2.6 M parameters, 6.2–7.6 GFLOPs (giga floating-point operations), and 22 ms latency on an NVIDIA RTX 4060 Laptop graphics processing unit (GPU), ensuring that result differences reflect architectural design rather than parameter or compute budget.

D. Experimental Configurations

1) *Axis 1 (RQ1) — Negative-Frame Ratio Sweep:* Holding the positive core fixed at $|\mathcal{D}_{\text{pos}}| = 2,401$ frames, we construct five nested training subsets at 20-percentage-point increments: 0% (2,401 total; 0 negative), 20% (3,001; 600 negative), 40% (4,001; 1,600 negative), 60% (6,003; 3,602 negative), and 80% (12,005; 9,604 negative). Subsets are strictly nested by increasing the negative pool with seed=42, so later splits are supersets of earlier ones. Axis 1 comprises 20 training runs (4 models × 5 ratios).

2) *Axis 2 (RQ2) — Hard-Negative Curation:* For each architecture, a positive-only baseline ($r=0\%$) runs inference over all 10,178 training negatives at $\tau=0.25$. False-alarm frames are ranked by maximum predicted confidence $s_{\text{max}}(X)$, and the top $N_{\text{target}}=1,600$ frames form the curated

TABLE II
FROZEN TRAINING HYPERPARAMETERS (ALL MODELS AND SPLITS)

Hyperparameter	Value
Input resolution	640 × 640 (FP32)
Batch / effective batch	16 / 16
Training duration	100 epochs
Optimizer	AdamW (lr ₀ =0.00125)
Weight decay	0.0005
Mosaic cooldown	Final 10 epochs
Random seed	42 (dual-seed: 42, 43 for $r \leq 40\%$)

Hyperparameters are identical across all architectures and ratio splits to isolate feature-aggregation design.

hard-negative set (backfilled deterministically from remaining negatives when needed). We evaluate curation at the reference ratio $r_{\text{ref}}=40\%$ with strictly matched cardinality: $|\mathcal{D}_{\text{train,hard}}^{(40\%)}| = |\mathcal{D}_{\text{train,rand}}^{(40\%)}| = 4,001$ frames, isolating sample difficulty from dataset volume. Axis 2 adds 8 training runs for a primary benchmark total of 28 runs.

E. Training Protocol

All models are fine-tuned from COCO (Common Objects in Context)-pretrained checkpoints at 640×640 resolution, single precision (FP32), using the AdamW optimizer (lr₀ = 0.00125, weight decay=0.0005, batch size=16) for 100 epochs with mosaic augmentation disabled for the final 10 epochs (close_mosaic=10). Table II summarizes the frozen hyperparameters, which are held identical across all models and ratio conditions.

Training-step note: Fixing epoch count over variable dataset sizes (2,401–12,005 frames) means the 80% condition receives approximately $5\times$ more gradient steps than the 0% condition. This confound is acknowledged; the convergence results at 80% should be interpreted as an upper bound on what extended training with high negative prevalence can achieve rather than a pure architectural effect.

IV. EXPERIMENTAL RESULTS

A. Ratio Sensitivity (RQ1)

Table III reports detection accuracy and operational false-alarm rates across all ratio-sweep conditions on the held-out test set (1,572 frames, 80.9% negative prevalence).

High-ratio convergence. At $r=0\%$, all models produce substantial false-alarm leakage because background frames generate no positive anchor assignments and thus no back-propagation gradients to suppress non-target activations. As negative prevalence scales to $r=80\%$, all four architectures converge to a narrow operational band (3.14–7.08 FP/1k). This convergence is driven by a strong monotonic inverse relationship between negative ratio and FP/1k (pooled Spearman $\rho = -0.937$, $p < 10^{-4}$; Pearson $r = -0.857$, $p < 10^{-4}$). Precision improves significantly with ratio (Spearman $\rho = +0.546$, $p < 0.05$), while foreground recall remains statistically uncorrelated ($\rho = +0.012$, $p > 0.50$), confirming that background regularization suppresses hallucinations without eroding target sensitivity.

TABLE III
NEGATIVE-FRAME RATIO SWEEP ACROSS ARCHITECTURES (RQ1)

Det.	r	mAP ₅₀	mAP ₅₀₋₉₅	P	R	FP/1k
YOLO11n	0%	0.9824	0.7570	0.8576	0.9787	67.22
	20%	0.9838	0.7524	0.9283	0.9814	31.05
	40%	0.9919	0.7711	0.9308	0.9831	22.01
	60%	0.9899	0.7723	0.9468	0.9806	9.83
	80%	0.9933	0.7606	0.9644	0.9766	4.72
YOLO26n	0%	0.9170	0.7086	0.8557	0.9105	80.59
	20%	0.9729	0.7546	0.9306	0.9763	16.90
	40%	0.9726	0.7388	0.9224	0.9519	9.43
	60%	0.9660	0.7594	0.9587	0.9086	10.22
	80%	0.9736	0.7671	0.9217	0.9794	3.14
YOLO12n	0%	0.9910	0.7496	0.9228	0.9823	45.20
	20%	0.9757	0.7449	0.8954	0.9863	35.77
	40%	0.9866	0.7659	0.9199	0.9949	18.86
	60%	0.9888	0.7766	0.9348	0.9960	9.43
	80%	0.9925	0.7529	0.9621	0.9767	7.08
YOLOv10n	0%	0.9423	0.7287	0.9065	0.8831	57.39
	20%	0.9423	0.7270	0.8833	0.9400	21.23
	40%	0.9642	0.7568	0.8813	0.9788	14.94
	60%	0.9749	0.7525	0.9221	0.9758	4.72
	80%	0.9524	0.7384	0.9010	0.9024	4.72

Pooled correlation with neg. ratio (N=20)

Spearman ρ	—	+0.546*	+0.012 ^{n.s.}	−0.937***
Pearson r	—	+0.628*	+0.147 ^{n.s.}	−0.857***

Dual-seed means (seed=42,43) for $r \leq 40\%$; single seed otherwise. FP/1k at $\tau=0.25$, IoU=0.70. P/R at max- F_1 ; 1,272 neg. test frames. Corr. block ($N=20$): pooled across models; * $p < 0.05$, *** $p < 10^{-4}$, n.s.: $p > 0.50$.

Bold: best per architecture.

Architectural divergence at intermediate ratios. The path to convergence differs sharply by paradigm (Fig. 1a). YOLO26n (RepConv) drops from 80.59 to 9.43 FP/1k by $r=40\%$ because localized, fused convolutional filters rapidly learn to suppress prominent cabin textures (steering-wheel rims, seatbelt buckles). In contrast, YOLO11n, YOLO12n, and YOLOv10n require $r=60\%$ – 80% to reach comparable suppression. Attention layers compute dense pairwise or area-based feature correlations; without sufficient background variety, they bind ambient cues such as windshield reflections and headrest contours into spurious foreground hypotheses. YOLOv10n’s dual-assignment architecture must simultaneously suppress over-assignment in both its auxiliary one-to-many head and its one-to-one inference head, requiring sustained negative supervision. The suppression divergence across architectures reaches $31.5\times$ (YOLO26n vs. YOLO12n) at $r=0\%$.

Foreground localization is preserved. mAP₅₀₋₉₅ remains within the range 0.738–0.777 across all ratio levels (Fig. 1b). Because negative frames carry no ground truth, bounding-box regression and distribution focal loss receive no update from them; only the classification branch tightens its decision boundary on background cells.

B. Hard-Negative Curation vs. Random Sampling (RQ2)

Table IV compares hard-negative mining against uniform random sampling at $r_{\text{ref}}=40\%$ with matched cardinality ($N=4,001$ frames). Values are dual-seed means (seed=42, 43).

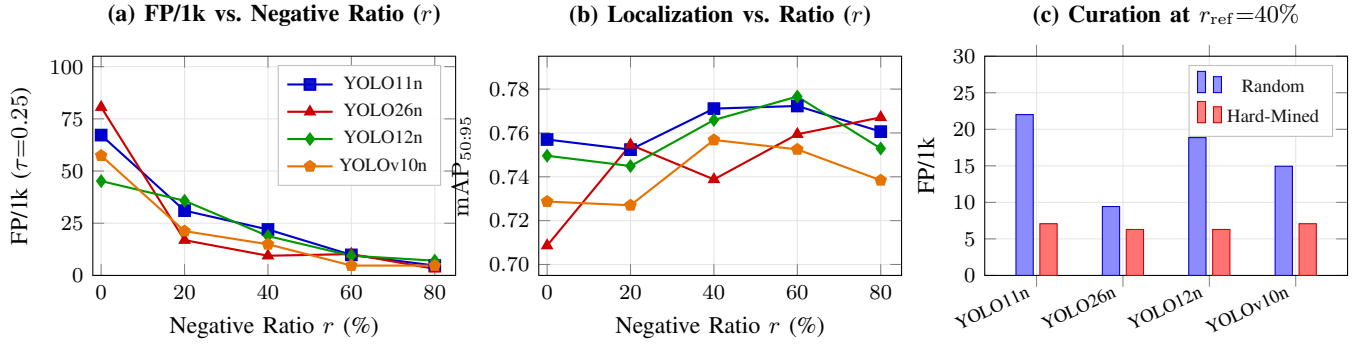


Fig. 1. Cross-architecture results. (a) FP/1k decreases monotonically as negative ratio r increases, converging to 3.14–7.08 FP/1k at $r=80\%$ across all architectures; suppression trajectories at intermediate ratios are paradigm-dependent. (b) $mAP_{50:95}$ is robust across all ratio levels (≥ 0.709), confirming that background training does not erode foreground localization. (c) At matched cardinality ($r_{\text{ref}}=40\%$, $N=4,001$), hard-negative curation cuts FP/1k by 33%–68% depending on architecture; attention and dual-assignment models gain most. Line and marker coding in (a) and (b) is shared.

TABLE IV
HARD-NEGATIVE MINING VS. RANDOM AT MATCHED CARDINALITY
($r_{\text{ref}}=40\%$, RQ2)

Detector	Curation	mAP_{50}	$mAP_{50:95}$	FP/1k	% Δ
YOLO11n	Random	0.9919	0.7711	22.01	
	Hard-Mined	0.9932	0.7738	7.08	−67.8%
YOLO26n	Random	0.9726	0.7388	9.43	
	Hard-Mined	0.9759	0.7518	6.29	−33.3%
YOLO12n	Random	0.9866	0.7659	18.86	
	Hard-Mined	0.9928	0.7839	6.29	−66.7%
YOLOv10n	Random	0.9642	0.7568	14.94	
	Hard-Mined	0.9788	0.7673	7.08	−52.6%

Evaluated on 1,272 negative test frames ($\tau=0.25$, matched $N=4,001$ training frames). % Δ : relative change in FP/1k (hard-mined vs. random). Bold marks best condition per architecture.

All architectures benefit from curation, but to different degrees. Hard-negative mining reduces FP/1k for every architecture; the magnitude of the gain is strongly architecture-dependent (Fig. 1c).

Attention and dual-assignment models (YOLO11n, YOLO12n, YOLOv10n): These models achieve 52.6%–67.8% false-alarm reductions alongside localization gains ($mAP_{50:95}$ increasing by up to +0.018). Uniform random sampling predominantly selects low-entropy background patches (plain headrests, uniform window glass) that deliver negligible gradient feedback to global query-key attention weights. Hard negatives—featuring complex cabin reflections and partial driver silhouettes—provoke spurious cross-attention activations, and explicit training on them forces attention layers to decorrelate these ambient patterns from true driver cues. For YOLOv10n, mined negatives penalize over-assignment in the auxiliary one-to-many head, sharpening the inference-time one-to-one branch.

Structural reparameterization (YOLO26n): Hard-negative mining still yields a meaningful 33.3% reduction ($9.43 \rightarrow 6.29$ FP/1k), but the absolute gain is smaller than for attention models. YOLO26n’s localized 3×3 RepConv kernels rapidly learn dominant cabin textures from even random negatives and saturate earlier; high-entropy hard negatives provide useful but

TABLE V
PROJECTED NUISANCE ALERTS PER HOUR ($\mathcal{A}_h$)

Detector	Training Split	5 FPS	15 FPS	30 FPS
YOLO11n	0% neg. baseline	870	2,610	5,220
	80% neg. ratio	69	206	412
YOLO26n	0% neg. baseline	1,443	4,328	8,655
	80% neg. ratio	46	137	274
YOLO12n	0% neg. baseline	756	2,267	4,534
	80% neg. ratio	103	309	619
YOLOv10n	0% neg. baseline	996	2,989	5,977
	80% neg. ratio	69	206	412

via (2) with $p_{\text{neg}}=0.809$. Bold: lowest alert rate per frame-rate column.

less incremental gradient signal because the model’s limited global receptive field cannot exploit long-range contextual distinctions.

Practical curation rule. The data indicate a clear architectural gradient in curation sensitivity: attention and dual-assignment architectures derive large gains from hard-negative mining at matched dataset size, while structural reparameterization models benefit more modestly. In resource-constrained settings where mining pipelines add engineering overhead, random sampling is adequate for RepConv-based models; for attention-based or dual-head models, the 52%–68% false-alarm reduction justifies the curation cost.

C. Operational Deployment Cost

Table V translates FP/1k values into projected hourly nuisance alert rates $\mathcal{A}_h$ via (2) at three standard streaming frame rates.

Without negative training, detectors trigger up to 8,655 nuisance alerts per hour at 30 FPS. Scaling negative frames to 80% cuts alert rates by up to 96.8% without changing a single model parameter. This analysis formalizes a key principle for edge AIoT systems: **inference latency is architecture-bound; operational reliability is curation-bound**. YOLOv10n achieves the fastest inference (7.7 ms, 129 FPS), but without negative training its alert rate (5,977/hr

at 30FPS) makes it impractical for continuous deployment. Data curation is the decisive design lever.

V. DISCUSSION

A. Practical Guidelines

The empirical results support two concrete recommendations for edge AIoT practitioners deploying DMS or similar continuous-sensing systems.

1) Prioritize negative-frame prevalence. Regardless of architecture, the single most impactful decision is ensuring that training data reflects operational negative prevalence. Training at $r=80\%$ negatives reduces FP/1k by $7.3\times$ – $25.7\times$ relative to the zero-negative baseline and converges all four architectures to a narrow operational band (3.14–7.08 FP/1k). For teams with sufficient data and compute, $r=80\%$ is the clear recommendation. For teams with constrained training budgets, $r=40\%$ already captures the bulk of this suppression (FP/1k ≤ 22 for all models) while using less than one-third the negative volume of the 80% split.

2) Match curation strategy to detector family. At a fixed training budget ($r=40\%$, $N=4,001$ frames), hard-negative mining reduces false alarms by 52.6%–67.8% for attention and dual-assignment models (YOLO11n, YOLO12n, YOLOv10n) and by 33.3% for the structural reparameterization model (YOLO26n). All four architectures benefit from curation, but the gain is proportional to the model’s capacity for global feature correlation. Teams deploying RepConv-based models can rely on simple random background sampling without a mining pipeline; teams deploying attention-based or dual-head models should invest in hard-negative curation to realize the larger available gains.

B. Limitations

- **Training-step confound.** Fixing epoch count over dataset sizes of 2,401–12,005 frames means the 80% condition receives approximately $5\times$ more gradient steps. Step-controlled ablations remain future work.
- **Mining purity variation.** Hard-mined candidate pools contained 34.2%–59.2% verified false-alarm frames (remainder: backfilled random negatives). The practical ceiling for curation gain therefore exceeds our reported figures.
- **Temporal correlation.** Consecutive frames share the same cabin interior and lighting. Near-ceiling mAP₅₀ > 0.97 may partly reflect within-vehicle rather than cross-vehicle generalization.
- **Foreground class imbalance.** The positive core is dominated by phone_use (81.2%), so aggregate mAP primarily tracks majority-class performance.
- **Single-platform latency.** Latency is profiled on an NVIDIA RTX 4060 Laptop GPU. Target embedded platforms such as the NVIDIA Jetson Orin Nano will show different absolute values and relative rankings.

VI. CONCLUSION

This paper investigated the joint effect of negative-frame prevalence and hard-negative curation on false-alarm suppression across four matched-capacity (2.3–2.6 M parameter) YOLO detectors in a continuous edge AIoT driver-monitoring task.

Three findings have immediate practical relevance. First, high negative prevalence ($r=80\%$) is the single most effective lever for suppressing false alarms: it reduces FP/1k by up to $25.7\times$ and drives all four architectures to a narrow operational band (3.14–7.08 FP/1k) without eroding foreground recall (Spearman $\rho = +0.012$, not significant). Second, suppression trajectories at intermediate ratios diverge substantially by paradigm: structural reparameterization models converge fast and early, while attention and dual-assignment models require higher negative densities. This means that detector selection and dataset curation are coupled design decisions, not independent ones. Third, hard-negative mining at matched cardinality yields large additional gains for attention and dual-assignment models (52.6%–67.8%) but a more modest gain for the RepConv model (33.3%), providing a concrete heuristic for when to invest in mining pipelines.

Together, these results shift the design-time question from “which architecture has the best mAP?” to “which architecture–curation combination achieves an acceptable hourly nuisance alert rate at my deployment frame rate?” Future work should control for gradient steps across ratio conditions, evaluate step-controlled schedules, extend the analysis to outdoor IoT sensing domains, and explore confidence-calibration methods that can close the remaining gap at the $r=40\%$ operating point.

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